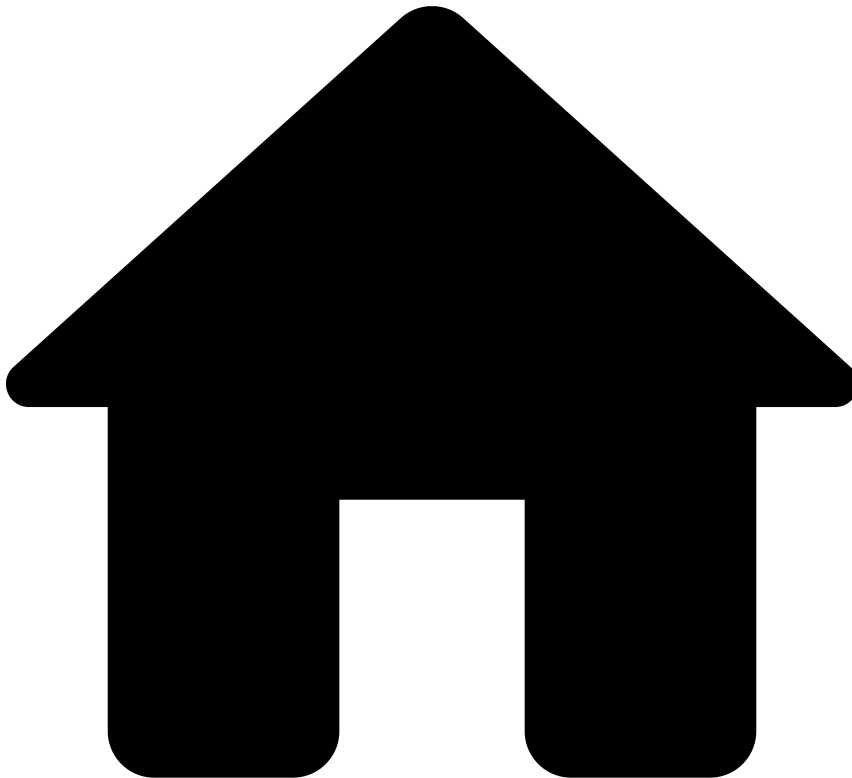


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On this page

Create an AutoML Run

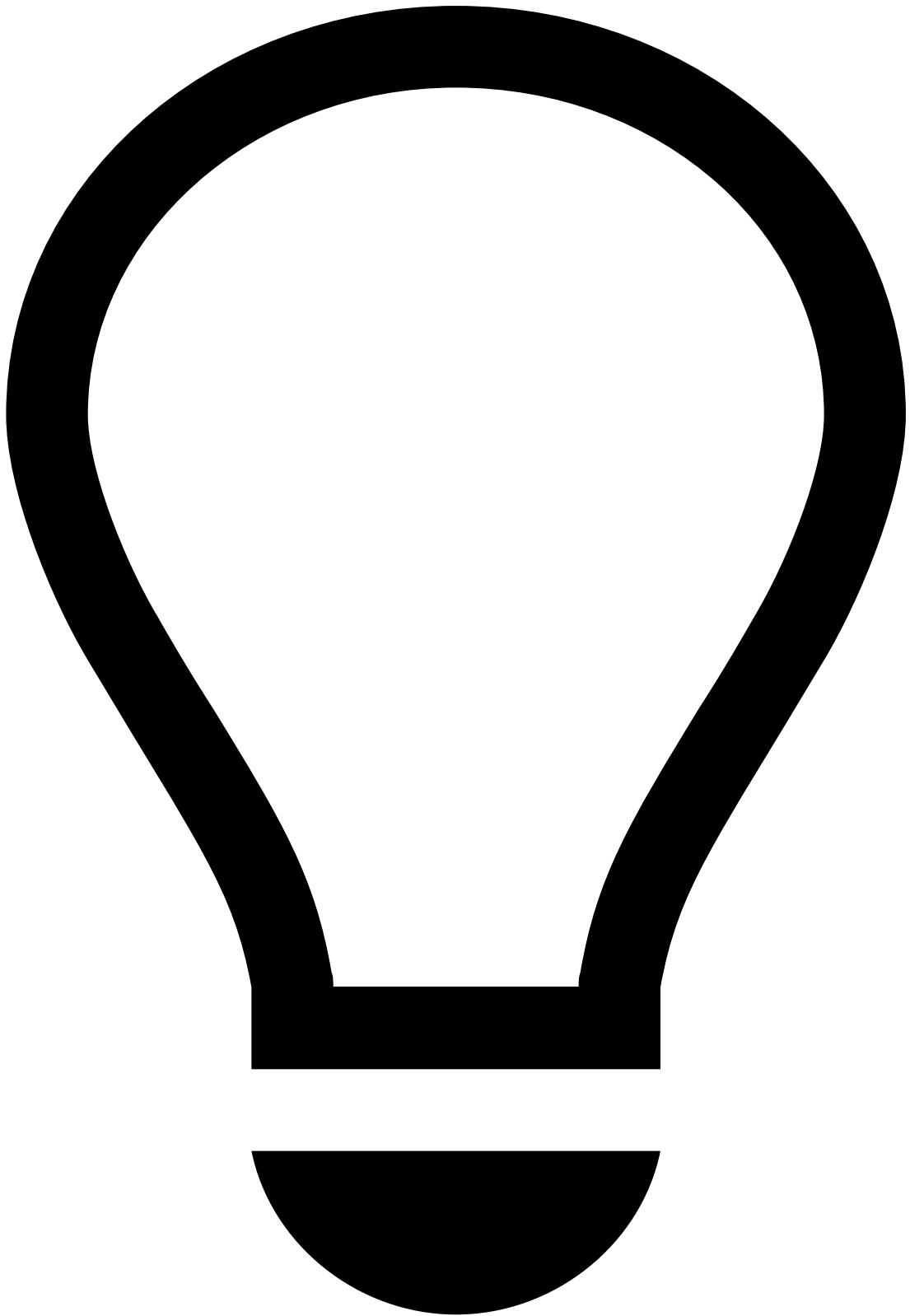
Was this page helpful? 

This page explains how to create an AutoML run to generate features and models automatically.



danger

Environment to run AutoML: Since the AutoML computation is quite resource-intensive for the system, Feedzai recommends running AutoML in a staging environment dedicated to the data science work (i.e. typically a Data Science environment). This is important to ensure that the production environment is not compromised by this functionality. Once you achieve results that meet your business needs, you can [migrate](#) the AutoML model and plan to the production environment and [integrate these new resources into the runtime workflow](#).



tip

Data Sampling: For large datasets, the computation of AutoML may stop the system. To prevent this, before creating an AutoML run, use a sampling strategy to remove part of the original dataset, while keeping a fair data representation.

Prerequisites:

Before running AutoML on a data source, you need to ensure that:

- The data values are valid. Check the complete list of data conversions in the [Prepare the data values - AutoML prerequisites](#) section.
- [Download](#) the semantic file that maps the data values to semantic tags that are relevant to your use case.

Create an AutoML run

To create an AutoML run that finds features and train models, perform the following actions:

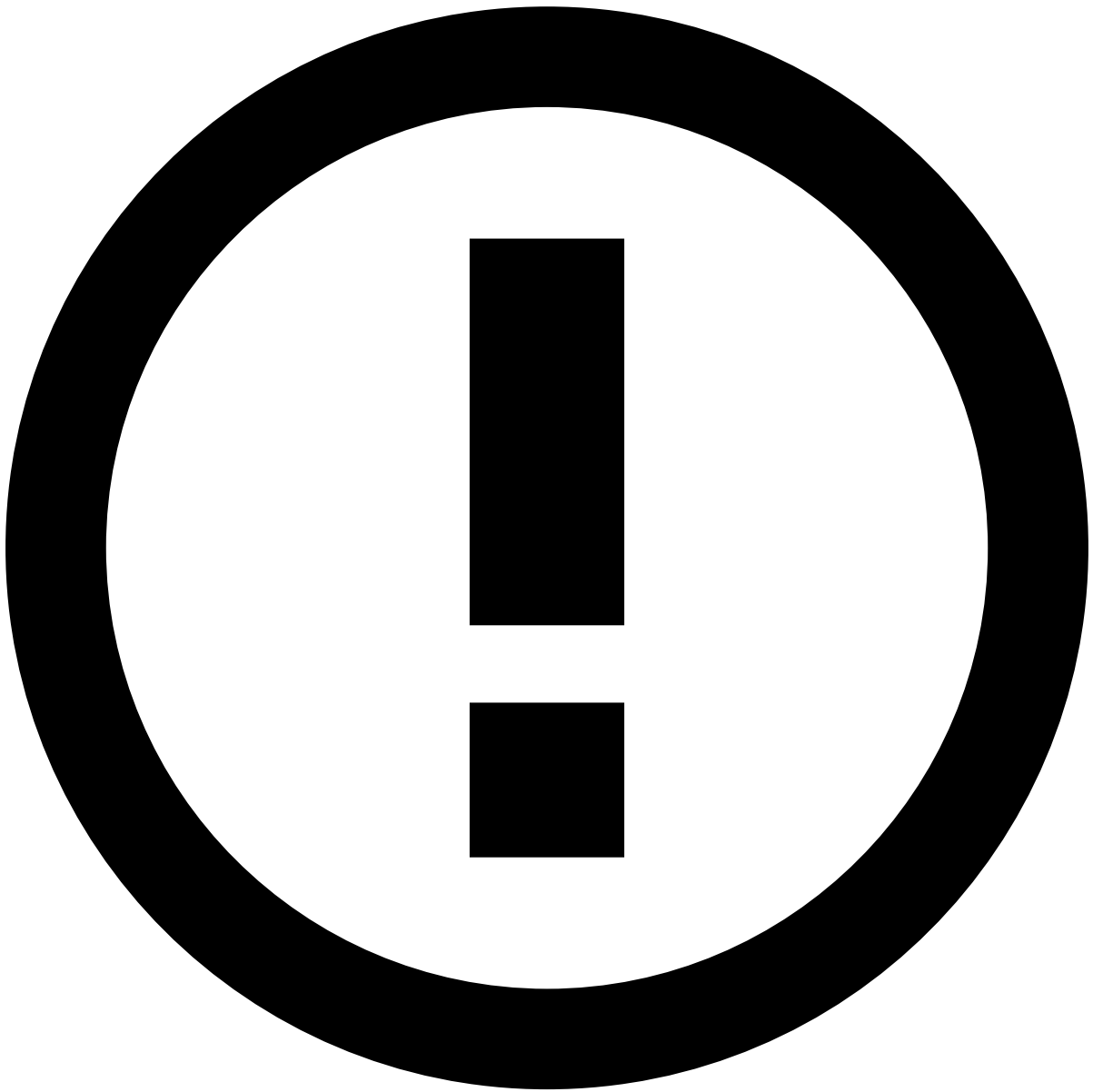
1. In the **Data Science** tab, expand the AutoML section using the button.
2. Select to start a new AutoML run.
3. Configure the data source and basic settings:
 1. Select the **Data Source** you will use to create features and train models.



info

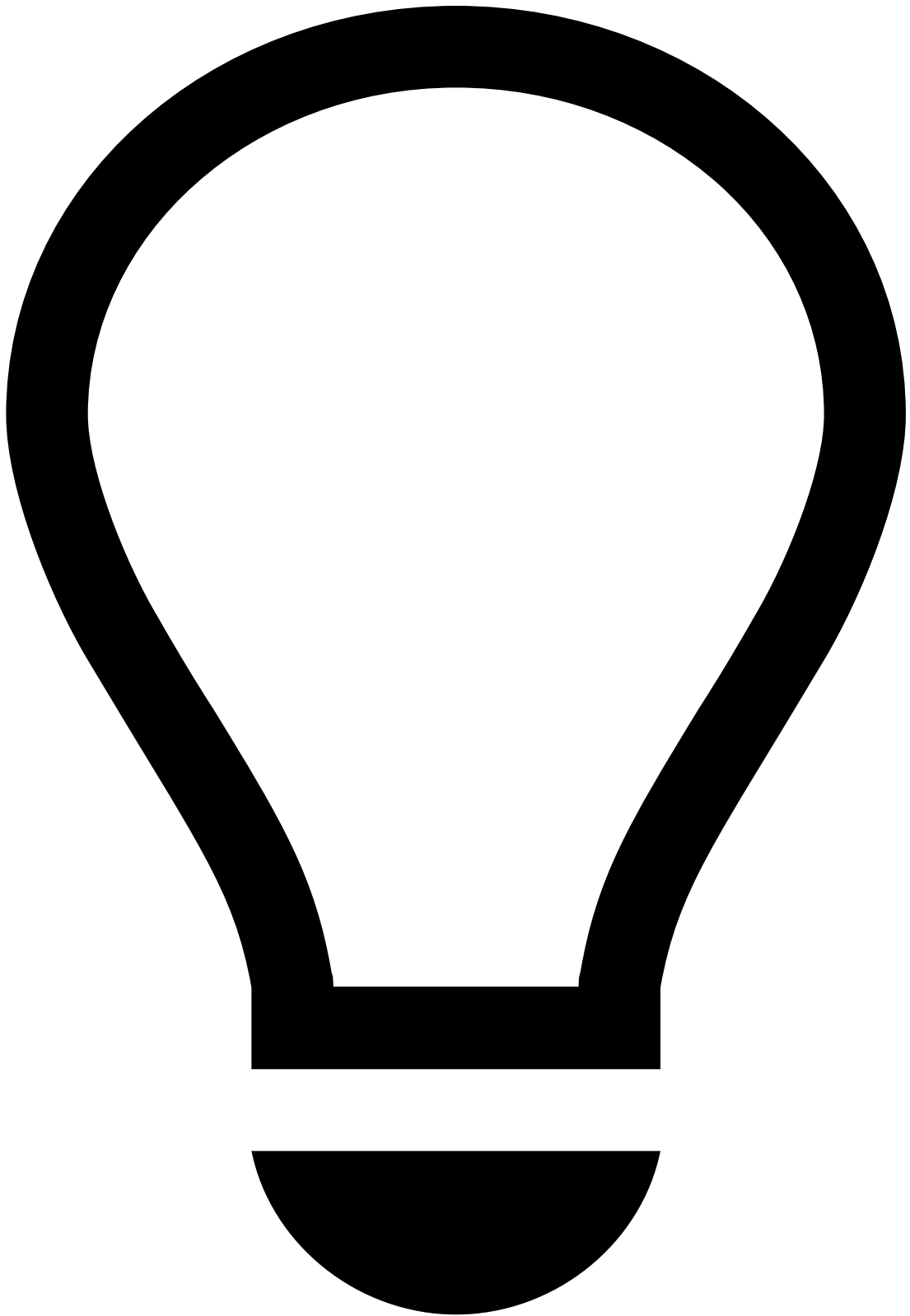
Data Source: If you want to integrate the recommended model that is generated by AutoML with the standard rules in the runtime workflow, make sure you select a data source that has the same schema instead of the one that is used by rules. Check the [Complete the Schema](#) page for more information about this step.

2. Upload the **Semantic File** that contains the AutoML tags for the data fields. Check the [AutoML Stages](#) page to learn more about the semantic file. The tags link the fields to their meaning. This file is provided by the standard Transaction Fraud for Banking package.
 3. Select the **Timestamp Field** from the list of available fields (e.g. `event_occurred_at`). Pulse uses the timestamp field to aggregate the data by time windows for the calculation of metrics.
 4. Select a categorical data field for the **Target Variable**. This is the variable that the models will predict, and is typically a field named as `fraud`.
 5. Select the **Target Value**, which is the value of the **Target Variable** that you want to detect. For example, in the fraud detection scenario, the **Target Variable** would be the `fraud` field, and the possible **Target Values** would be 'True' and 'False'. To detect fraud, select 'True' for the **Target Value**.
 6. Enable an **Additional Cost Field** for model evaluations. Select a field that represents the transaction amount to have metrics such as the Money Recall and Money False Positive Rate (FPR).
 7. Select the option **Next** to proceed to the next stage.
4. Split the data source into smaller datasets for training, validation and evaluation. To do so, define the following:
1. **Time range for training:** The training dataset will be used for training models during the hyperparameter tuning phase of the AutoML run.
 2. **Time range for validation:** The validation dataset will be used to evaluate the models during the hyperparameter tuning phase to find the best model.
 3. Use the **Enable test dataset** switch to define the time range for the dataset that will be used for the final evaluation of the best model found. If you don't define a test dataset, Pulse will use the validation dataset for this purpose.



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Time ranges: In time ranges, the beginning timestamp is inclusive, while the ending timestamp is exclusive.



tip

Planning training, validation and test data sources: To avoid under- or overfitting, plan the time ranges carefully. Ensure that the resulting datasets contain enough data for their purpose. A typical dataset split is 50% for training, 30% for validation and 20% for evaluation, as long as the data source size is sufficient for that. The time windows need to be in the past and follow the order training-validation-evaluation.

If you don't define a test dataset, AutoML uses the validation dataset for the final evaluation of the best model found. Since the validation dataset was used for finding the best model, it is not an optimal solution to use it to test the data. Therefore, you should use a test dataset.

1. Select **Metric to optimize**, and a fixed metric value that you want to achieve. For example, you may want to optimize your Recall as your metric while ensuring an FPR around 1% as your fixed metric value.
2. You can configure **Advanced options** as follows:

1. Set the **Number of Models** you want to train in this AutoML run.



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Computational weight: This value heavily influences the runtime of the AutoML job. Feedzai recommends that you test the runtime with a few models first.

2. Select the **Path to generated data source** if you want to save the data in a specific path.
3. Select the **Window size** for metrics. If the window size of a metric is 1 hour, it means that the metric calculated for a certain point in time is based on the data of the last hour before that given time. Each defined window size will generate another metric.
 - Metrics with **Real-time Updates** will be updated in every event.
 - Metrics with **Scheduled Updates** will be updated at the start of each day.
3. Select **Create** to start the AutoML job.

Follow up the AutoML run¹

After setting up the AutoML process, you can follow up on its status at the top of the **Data Science** tab. When the AutoML job is finished, you can view the results in the same place:

- The **Best Model** trained in the last AutoML run.
- The **Features Plan** that created a data source with features based on Feedzai's experience in fraud detection.
- The **Details** about the job that executed the last AutoML run. If the AutoML run fails, you can start investigating [here](#).

Next steps²

Once the AutoML job has finished, [analyze the AutoML Output](#). One of the key outputs of an AutoML run is the recommended model. The next page helps you analyze the performance of the model and provides details about additional resources produced by the AutoML such as plans and data sources.